

Experiment – 12
CSA1717 – Artificial Intelligence

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Aim:

To write a Python program to add two matrices of size $m \times n$ and display the resultant matrix.

Algorithm:

1. Start the program.
2. Read the number of rows and columns of the matrices.
3. Create Matrix A and read its elements.
4. Create Matrix B and read its elements.
5. Initialize an empty matrix C to store the result.
6. For each position (i, j):

Add the corresponding elements:

$$C[i][j] = A[i][j] + B[i][j]$$

7. Display Matrix C.
8. Stop the program.

Source Code:

```
# Program to add two matrices

# Taking matrix size
rows = int(input("Enter number of rows: "))
cols = int(input("Enter number of columns: "))

# Input Matrix A
print("\nEnter elements of Matrix A:")
A = []
for i in range(rows):
```

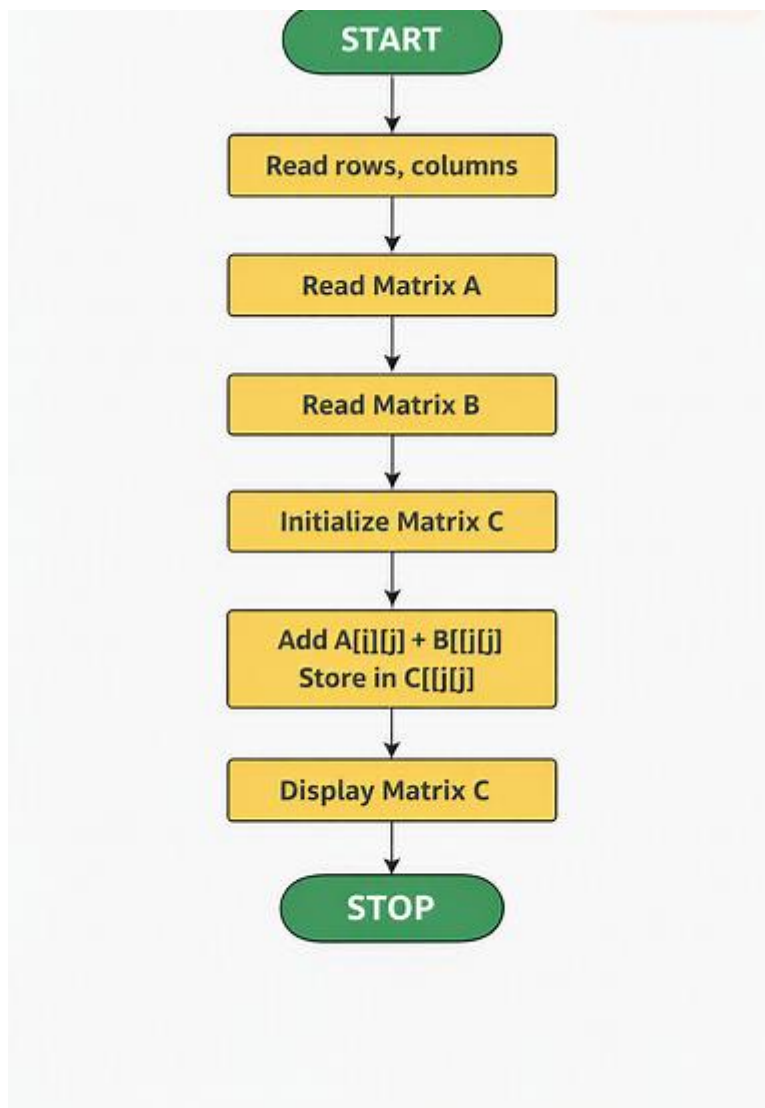
```
row = []  
for j in range(cols):  
    row.append(int(input(f'A[{i}][{j}]: ')))  
A.append(row)
```

```
# Input Matrix B  
print("\nEnter elements of Matrix B:")  
B = []  
for i in range(rows):  
    row = []  
    for j in range(cols):  
        row.append(int(input(f'B[{i}][{j}]: ')))  
    B.append(row)
```

```
# Adding matrices  
C = []  
for i in range(rows):  
    row = []  
    for j in range(cols):  
        row.append(A[i][j] + B[i][j])  
    C.append(row)
```

```
# Display Result  
print("\nResultant Matrix (A + B):")  
for i in range(rows):  
    for j in range(cols):  
        print(C[i][j], end=" ")  
    print()
```

Flowchart:



Output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d,  
Enter "help" below or click "Help" a  
>>>  
= RESTART: C:/Users/hemal/OneDrive/D  
Enter number of rows: 2  
Enter number of columns: 2  
  
Enter elements of Matrix A:  
A[0][0]: 1  
A[0][1]: 2  
A[1][0]: 3  
A[1][1]: 4  
  
Enter elements of Matrix B:  
B[0][0]: 5  
B[0][1]: 6  
B[1][0]: 7  
B[1][1]: 8  
  
Resultant Matrix (A + B):  
6 8  
10 12
```

Result:

The program successfully reads two matrices, adds their corresponding elements, and displays the resultant matrix.